Mechanics

May 26, 2025

1 Basic Elementary Principles

Result 1 *(Newton's law)*

1. In an inertial reference frame, an object remains at rest or constant velocity unless acted upon by external force.
2. $$\dot{\boldsymbol{p}} = \frac{d(m\boldsymbol{v})}{dt} = \boldsymbol{F}^{(e)}$$
3. Action and reaction: $\boldsymbol{F}_{ij} = -\boldsymbol{F}_{ji}$ (Additional condition for strong version: $\boldsymbol{r}_{ij} \times \boldsymbol{F}_{ji} = 0$)

1.1 Single particle

Result 2 (Conservation of Linear Momentum)

$$\textit{If } \boldsymbol{F} = 0, \textit{ then } \dot{\boldsymbol{p}} = 0$$

Angular momentum of the particle around point O is $\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p}$. Torque $\boldsymbol{N} = \boldsymbol{r} \times \boldsymbol{F}$.

$$\boldsymbol{N} = \frac{d}{dt}(\boldsymbol{r} \times m\boldsymbol{v}) = \frac{d\boldsymbol{L}}{dt} \equiv \dot{\boldsymbol{L}}$$

Result 3 (Conservation of Angular Momentum)

$$\text{If } \boldsymbol{N} = 0, \text{ then } \dot{\boldsymbol{L}} = 0$$

If force field $\boldsymbol{F}$ is conservative ($\oint \boldsymbol{F} \cdot d\boldsymbol{s} = 0$ or $\int_a^b \boldsymbol{F} \cdot d\boldsymbol{s} = T_b - T_a$ is the same for any path between a and b). There exists a potential scalar field V such that

$$\boldsymbol{F} = -\boldsymbol{\nabla} V(\boldsymbol{r})$$

Result 4 (Conservation of Energy)

$E = T + V$ is conserved where T and V are kinetic energy and potential energy respectively

1.2 System of particles

Center of mass:

$$\boldsymbol{R} = \frac{\sum_i m_i \boldsymbol{r}_i}{\sum m_i} = \frac{\sum_i m_i \boldsymbol{r}_i}{M}$$

Let $\boldsymbol{F}_i^{(e)}$ external force acting on particle i-th, and $\boldsymbol{F}_{ji}$ is the force exerted by j-th particle on i-th particle in the system,

$$m_i \frac{d^2(\boldsymbol{r_i})}{dt^2} = \dot{\boldsymbol{p}}_{\boldsymbol{i}} = \sum_j \boldsymbol{F}_{ji} + \boldsymbol{F}_i^{(e)}$$

Summing over all particles, we get

$$\dot{\boldsymbol{p}} = M \frac{d^2 \boldsymbol{R}}{dt^2} = \sum_i \boldsymbol{F}_i^{(e)} \equiv \boldsymbol{F}^{(e)}$$

Result 5 (Conservation of Linear Momentum for a system of particles)

if the total external force $\boldsymbol{F}^{(e)} = 0$, the total linear momentum $\boldsymbol{p}$ is conserved.

The change of total angular momentum:

$$\begin{aligned}
\frac{d}{dt}\boldsymbol{L} &= \sum_i \boldsymbol{r_i} \times \dot{\boldsymbol{p}}_{\boldsymbol{i}} = \sum_i \boldsymbol{r_i} \times (\boldsymbol{F}_i^{(e)} + \sum_j \boldsymbol{F}_{ji}) \\
&= \boldsymbol{N}^{(e)} + \sum_{i,j} \boldsymbol{r_i} \times \boldsymbol{F}_{ji} \\
&= \boldsymbol{N}^{(e)} + \sum_{i<j} \boldsymbol{r_i} \times \boldsymbol{F}_{ji} + \sum_{j<i} \boldsymbol{r_i} \times \boldsymbol{F}_{ji} \\
&= \boldsymbol{N}^{(e)} + \sum_{i<j} \boldsymbol{r_i} \times \boldsymbol{F}_{ji} + \sum_{i<j} \boldsymbol{r_j} \times \boldsymbol{F}_{ij} \\
&= \boldsymbol{N}^{(e)} + \sum_{i<j} \boldsymbol{r_i} \times \boldsymbol{F}_{ji} - \sum_{i<j} \boldsymbol{r_j} \times \boldsymbol{F}_{ji} \\
&= \boldsymbol{N}^{(e)} + \sum_{i<j} (\boldsymbol{r_i} - \boldsymbol{r_j}) \times \boldsymbol{F}_{ji} \\
&= \boldsymbol{N}^{(e)} + \sum_{i<j} \boldsymbol{r}_{ij} \times \boldsymbol{F}_{ji}
\end{aligned}$$

When the internal force abides strong version of Newton's 3rd law, internal force pair has the same direction as the two particles $\boldsymbol{r}_{ij}$. Then $\boldsymbol{r}_{ij} \times \boldsymbol{F}_{ji} = 0$. Forces satisfy strong Newton's 3rd law are **central**.

Result 6 (Conservation of Angular Momentum for a system of particles)

$\boldsymbol{L}$ is constant in time if applied external torque $\boldsymbol{N}^{(e)} = 0$ and internal forces are central

Let $\boldsymbol{r}_i = \boldsymbol{R} + \boldsymbol{r}_i'$ where $\boldsymbol{r}_i'$ is the position of ith particle wrt to $\boldsymbol{R}$ instead of O. Similarly, $\boldsymbol{v}_i = \boldsymbol{v} + \boldsymbol{v}_i'$ where $\boldsymbol{v} = \dot{\boldsymbol{R}}$. We can find $\boldsymbol{R} = \frac{\sum_i m_i \boldsymbol{r}_i}{M} = \frac{\sum_i m_i (\boldsymbol{R} + \boldsymbol{r}_i')}{M} = \boldsymbol{R} + \sum_i \boldsymbol{r}_i'$ which means

$$\sum_i \boldsymbol{r}_i' = 0, \qquad \sum_i \dot{\boldsymbol{r}}_i' = \sum_i \boldsymbol{v}_i' = 0$$

.

Result 7 The total angular momentum about O is the angular momentum of center of mass + angular momentum of each particle around the center of mass.

$$\begin{aligned}\boldsymbol{L} &= \sum_i \boldsymbol{r_i} \times \boldsymbol{p_i} \\ &= \boldsymbol{R} \times M\boldsymbol{v} + \sum_i \boldsymbol{r}'_i \times \boldsymbol{p}'_i + \sum_i \boldsymbol{R} \times m_i \boldsymbol{v}'_i + \sum_i m_i \boldsymbol{r}'_i \times \boldsymbol{v} \\ &= \boldsymbol{R} \times M\boldsymbol{v} + \sum_i \boldsymbol{r}'_i \times \boldsymbol{p}'_i\end{aligned}$$

In general $\boldsymbol{L}$ depends on choice of O through $\boldsymbol{R}$ except when the center of mass is at rest $\boldsymbol{R} = 0$.

Result 8 The total kinetic energy of the system is

$$T = \sum_i \frac{1}{2} m_i v_i^2 = \sum_i \frac{1}{2} m_i (\boldsymbol{v} + \boldsymbol{v}'_i)(\boldsymbol{v} + \boldsymbol{v}'_i) = \frac{1}{2} M v^2 + \sum_i \frac{1}{2} m_i (v'_i)^2$$

The ith particle has trajectory $\boldsymbol{r}_i(t)$ from time a to b. Then the work is the change in kinetic energy.

$$W_i = \int_a^b \boldsymbol{F}_i \cdot d\boldsymbol{r_i} = \int_a^b \frac{d(m_i v_i)}{dt} \cdot \frac{d\boldsymbol{r_i}}{dt} dt = \int_a^b m_i \dot{\boldsymbol{v}}_i \cdot \boldsymbol{v}_i dt = \int_a^b d(\frac{1}{2} m_i v_i^2) = T_i(b) - T_i(a)$$

Also we can write

$$W_i = \int_a^b \boldsymbol{F}_i \cdot d\boldsymbol{r_i} = \int_a^b \boldsymbol{F}_i^{(e)} \cdot d\boldsymbol{r_i} + \sum_j \int_a^b \boldsymbol{F}_{ji} \cdot d\boldsymbol{r_i}$$

If the external force $\boldsymbol{F}_i^{(e)}$ is conservative, then $\boldsymbol{F}_i^{(e)} = -\boldsymbol{\nabla} V_i$

$$\int_a^b \boldsymbol{F}_i^{(e)} \cdot d\boldsymbol{r_i} = -\int_a^b \boldsymbol{\nabla} V_i \cdot d\boldsymbol{r_i} = -\int_a^b \frac{\partial V_i}{\partial x_i} dx_i + \frac{\partial V_i}{\partial y_i} dy_i + \frac{\partial V_i}{\partial z_i} dz_i = -\int_a^b dV_i = -V_i|_a^b$$

If the action/reaction forces are conservative and central, $V_{ji}(|\boldsymbol{r_i} - \boldsymbol{r_j}|)$ only depends on distance between ith and jth particle, hence it is symmetric $V_{ji} = V_{ij}$. Also

$$\boldsymbol{F}_{ji} = -\boldsymbol{\nabla}_i V_{ij} = -\boldsymbol{F}_{ij} = \boldsymbol{\nabla}_j V_{ji} = \boldsymbol{\nabla}_j V_{ij}$$

We can find the gradients of V_{ij} with respect to $\boldsymbol{r_{ij}} = \boldsymbol{r_i} - \boldsymbol{r_j}$,

$$\boldsymbol{\nabla}_i V_{ij} = \frac{\partial V_{ij}}{\partial \boldsymbol{r_i}} = \frac{\partial V_{ij}}{\partial \boldsymbol{r_{ij}}} \frac{\partial \boldsymbol{r_{ij}}}{\partial \boldsymbol{r_i}} = \frac{\partial V_{ij}}{\partial \boldsymbol{r_{ij}}} = \boldsymbol{\nabla}_{ij} V_{ij} = -\boldsymbol{\nabla}_j V_{ij}$$

The work done by internal forces can be written in sums of pairs of potentials between ith and jth particle.

$$\begin{aligned}
\sum_{i,j}\int_a^b \boldsymbol{F}_{ji}\cdot d\boldsymbol{r_i} &= -\sum_{i,j}\int_a^b \boldsymbol{\nabla}_j V_{ji}\cdot d\boldsymbol{r_i} \\
&= -\sum_{i<j}\int_a^b \boldsymbol{\nabla}_j V_{ji}\cdot d\boldsymbol{r_i} + \boldsymbol{\nabla}_i V_{ij}\cdot d\boldsymbol{r_j} \\
&= -\sum_{i<j}\int_a^b \boldsymbol{\nabla}_j V_{ij}\cdot d\boldsymbol{r_i} - \boldsymbol{\nabla}_j V_{ij}\cdot d\boldsymbol{r_j} \\
&= -\sum_{i<j}\int_a^b \boldsymbol{\nabla}_j V_{ij} d(\boldsymbol{r_i}-\boldsymbol{r_j}) \\
&= -\sum_{i<j}\int_a^b \boldsymbol{\nabla}_j V_{ij} d(\boldsymbol{r_{ij}}) \\
&= \sum_{i<j}\int_a^b \boldsymbol{\nabla}_{ij} V_{ij} d(\boldsymbol{r_{ij}}) \qquad \text{(note that } \boldsymbol{\nabla}_{ij}V_{ij} = -\boldsymbol{\nabla}_j V_{ij}\text{)} \\
&= \frac{1}{2}\sum_{i,j}\int_a^b \boldsymbol{\nabla}_{ij} V_{ij} d(\boldsymbol{r_{ij}}) \\
&= \frac{1}{2}\sum_{i,j} V_{ij}|_a^b
\end{aligned}$$

Result 9 Total potential energy V when external and internal forces are conservative and internal forces also central is:

$$V = \sum_i V_i + \frac{1}{2}\sum_{i,j} V_{ij}$$

Such that total energy $T+V$ is conserved

On a rigid body, $|\boldsymbol{r_{ij}}|$ is constant between any two points. Then $0 = d(r_{ij}^2) = d(\boldsymbol{r}_{ij}\cdot\boldsymbol{r}_{ij}) = 2\boldsymbol{r}_{ij}\cdot d(\boldsymbol{r}_{ij})$. Therefore $d(\boldsymbol{r}_{ij})$ is perpendicular to $\boldsymbol{r}_{ij}$ which means the work done by internal force $\boldsymbol{F}_{ji}\cdot d(\boldsymbol{r}_{ij}) = 0$

1.3 Constraints

Holonomic Constraints have the form

$$f(\boldsymbol{r_1}, \boldsymbol{r_2}, \boldsymbol{r_3}, \ldots, t) = 0$$

For rigid body, $r_{ij}^2 - c_{ij}^2 = 0$. Constraints can't be expressed as above are called non-holonomic (e.g. placing a particle on a sphere surface is non-holonomic: $r^2 - a^2 \geq 0$, it will slide and eventually fall off the sphere).

Rheonomous Constraints contains time as an explicit variable. **Scleronomous Constraints** are not explicitly dependent on time.

Constraints difficulties:

1. the coordinates r_i are no longer independent.
2. forces of constraint are unknown a priori but we know the effect. e.g. force of wall on gas particle.

Generalized Coordinates can solve problem 1. Consider a system of N particles without constraints, it has $3N$ independent coordinate (**degrees of freedom**). When there are K **holonomic** constraints, we only have $3N-K$

degree of freedom, so we can re-parameterize the system with $q_1, \ldots, q_{3N-K}$ independent variables.

$$\boldsymbol{r}_i = \boldsymbol{r}_i(q_1, q_2, \ldots, q_{3N-k}, t), \quad \text{for } 1 \leq i \leq N$$

If the constraints are non-holonomic, we cannot eliminate the dependent variables this way.

1.4 D'Alembert Principle and Lagrangian equations

virtual displacement of a system is a change of the configuration of the system as the result of any arbitrary infinitesimal change of the coordinates: $\delta \boldsymbol{r}_i$, *consistent with the forces and constraints imposed on the system at the given instant* t.

This is different from actual displacement of a system in time interval dt during which the force and constraints might be changing.

When system is in equilibrium, total force on each partcile $\boldsymbol{F}_i = 0$. So the **virtual work** $\boldsymbol{F}_i \cdot \delta \boldsymbol{r}_i = 0$. Note that total force is the sum of applied force and constraint force: $\boldsymbol{F_i} = \boldsymbol{F}_i^{(a)} + \boldsymbol{f}_i$. Therefore

$$\sum_i \boldsymbol{F}_i \cdot \delta \boldsymbol{r}_i = \sum_i \boldsymbol{F}_i^{(a)} \cdot \delta \boldsymbol{r}_i + \sum_i \boldsymbol{f}_i \cdot \delta \boldsymbol{r}_i = 0$$

Result 10 (Principle of virtual work) We assume **the total net virtual work done by constraint forces are zero** which is $\sum_i \boldsymbol{f}_i \cdot \delta \boldsymbol{r}_i = 0$, we get the principle of virtual work for equilibrium for a system:

$$\sum_i \boldsymbol{F}_i^{(a)} \cdot \delta \boldsymbol{r}_i = 0$$

In general, $\boldsymbol{F}_i^{(a)} \neq 0$ and $\delta \boldsymbol{r}_i$ are not independent, so they can't vary freely. Not much can be said about them.

Note that $\boldsymbol{F}_i = \dot{\boldsymbol{p}}_i \Rightarrow \boldsymbol{F}_i - \dot{\boldsymbol{p}}_i = 0$, we can write

$$\begin{aligned}
&\sum_i (\boldsymbol{F}_i - \dot{\boldsymbol{p}}_i) \cdot \delta \boldsymbol{r}_i = 0 \\
\Rightarrow &\sum_i (\boldsymbol{F}_i^{(a)} - \dot{\boldsymbol{p}}_i + \boldsymbol{f}_i) \cdot \delta \boldsymbol{r}_i = 0 \\
\Rightarrow &\sum_i (\boldsymbol{F}_i^{(a)} - \dot{\boldsymbol{p}}_i) \cdot \delta \boldsymbol{r}_i + \boldsymbol{f}_i \cdot \delta \boldsymbol{r}_i = 0 \\
\Rightarrow &\sum_i (\boldsymbol{F}_i^{(a)} - \dot{\boldsymbol{p}}_i) \cdot \delta \boldsymbol{r}_i = 0
\end{aligned}$$

Result 11 (D'Alembert Principle) For an equilibrium system where the total net virtual work done by constraint forces are zero, we have

$$\sum_i (\boldsymbol{F}_i^{(a)} - \dot{\boldsymbol{p}}_i) \cdot \delta \boldsymbol{r}_i = 0$$

Suppose $\boldsymbol{r}_i = \boldsymbol{r}_i(q_1, \ldots, q_n, t)$, we have velocity,

$$\boldsymbol{v}_i = \dot{\boldsymbol{r}}_i = \sum_j \frac{\partial \boldsymbol{r}_i}{\partial q_j} \dot{q}_j + \frac{\partial \boldsymbol{r}_i}{\partial t}$$

and

$$\delta \boldsymbol{r}_i = \sum_j \frac{\partial \boldsymbol{r_i}}{\partial q_j} \delta q_j$$

Note that

$$\frac{\partial \boldsymbol{v}_i}{\partial \dot{q}_k} = \frac{\partial \boldsymbol{r}_i}{\partial q_j}\delta_j^k = \frac{\partial \boldsymbol{r}_i}{\partial q_k}, \qquad \frac{d}{dt}\left(\frac{\partial \boldsymbol{r}_i}{\partial q_j}\right) = \sum_k \frac{\partial^2 \boldsymbol{r}_i}{\partial q_j \partial q_k}\dot{q}_k + \frac{\partial^2 \boldsymbol{r}_i}{\partial q_j \partial t} = \frac{\partial \boldsymbol{v}_i}{\partial q_j} = \frac{\partial \dot{\boldsymbol{r}}_i}{\partial q_j}$$

Then we have the following:

$$\begin{aligned}
&\sum_i \dot{\boldsymbol{p}}_i \cdot \delta \boldsymbol{r}_i \\
&= \sum_i (m_i \dot{\boldsymbol{v}}_i) \cdot \sum_j \frac{\partial \boldsymbol{r_i}}{\partial q_j}\delta q_j \\
&= \sum_j \sum_i m_i \frac{d\boldsymbol{v}_i}{dt} \cdot \frac{\partial \boldsymbol{v_i}}{\partial \dot{q}_j}\delta q_j \\
&= \sum_j \left\{ \sum_i \left(m_i \frac{d\boldsymbol{v}_i}{dt} \cdot \frac{\partial \boldsymbol{v_i}}{\partial \dot{q}_j} + m_i \boldsymbol{v}_i \cdot \frac{d}{dt}\frac{\partial \boldsymbol{v_i}}{\partial \dot{q}_j} \right) - m_i \boldsymbol{v}_i \cdot \frac{d}{dt}\frac{\partial \boldsymbol{v_i}}{\partial \dot{q}_j} \right\} \delta q_j, \quad \text{(adding 0)} \\
&= \sum_j \left\{ \sum_i \frac{d}{dt}\left(m_i \boldsymbol{v}_i \cdot \frac{\partial \boldsymbol{v}_i}{\partial \dot{q}_j} \right) - m_i \boldsymbol{v}_i \cdot \frac{\partial \boldsymbol{v}_i}{\partial q_j} \right\} \delta q_j, \quad \text{(chain rule)} \\
&= \sum_j \left\{ \sum_i \frac{d}{dt}\left(\frac{1}{2} m_i \frac{\partial (\boldsymbol{v}_i \cdot \boldsymbol{v}_i)}{\partial \dot{q}_j} \right) - \frac{1}{2} m_i \frac{\partial (\boldsymbol{v}_i \cdot \boldsymbol{v}_i)}{\partial q_j} \right\} \delta q_j \\
&= \sum_j \left\{ \sum_i \frac{d}{dt}\left(\frac{\partial T_i}{\partial \dot{q}_j} \right) - \frac{\partial T_i}{\partial q_j} \right\} \delta q_j, \quad (T_i \text{ is the kinetic energy } \frac{1}{2} m_i v_i^2) \\
&= \sum_j \left\{ \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} \right\} \delta q_j
\end{aligned}$$

We define **Generalized force** Q_j such that

$$\sum_i \boldsymbol{F}_i \cdot \delta \boldsymbol{r}_i = \sum_j \left(\sum_i \boldsymbol{F}_i \cdot \frac{\partial \boldsymbol{r}_i}{\partial q_j} \right) \delta q_j \equiv \sum_j Q_j \delta q_j$$

Result 12 Then the D'Alember principle equation becomes

$$\begin{aligned}
&\sum_i (\boldsymbol{F}_i^{(a)} - \dot{\boldsymbol{p}}_i) \cdot \delta \boldsymbol{r}_i = 0 \\
\Rightarrow &\sum_j \left\{ \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} - Q_j \right\} \delta q_j = 0 \\
\Rightarrow &\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} - Q_j = 0 \\
\Rightarrow &\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} = Q_j
\end{aligned}$$

Note: because each δq_j is independent of one another, so the coefficient must vanish.

The term $\frac{\partial T}{\partial q_j}$ is due to the curvature of the generalized coordinate q_j (would be 0 in Cartesian coordinate).

Now suppose the applied forces are conservatives which means $\boldsymbol{F}_i = -\boldsymbol{\nabla} V_i$, then

$$Q_j = \sum_i \boldsymbol{F}_i \cdot \frac{\partial \boldsymbol{r}_i}{\partial q_j} = -\sum_i \boldsymbol{\nabla} V_i \cdot \frac{\partial \boldsymbol{r}_i}{\partial q_j} = -\sum_{i,k} \frac{\partial V_i}{\partial r_i^k}\frac{\partial r_i^k}{\partial q_j} \equiv -\frac{\partial V}{\partial q_j}$$

Where $V(\boldsymbol{r}_1, \boldsymbol{r}_2, \ldots, \boldsymbol{r}_N, t)$.

Then

$$\begin{aligned}
&\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_j}\right) - \frac{\partial T}{\partial q_j} - Q_j = 0 \\
\Rightarrow &\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_j}\right) - \frac{\partial T}{\partial q_j} + \frac{\partial V}{\partial q_j} = 0 \\
\Rightarrow &\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_j}\right) - \frac{\partial (T-V)}{\partial q_j} = 0 \\
\Rightarrow &\frac{d}{dt}\left(\frac{\partial (T-V)}{\partial \dot{q}_j}\right) - \frac{\partial (T-V)}{\partial q_j} = 0, \quad \text{(Because } \frac{\partial V}{\partial \dot{q}_j} = 0 \text{)}
\end{aligned}$$

Result 13 (Lagrange's equation) If the constraints forces do zero net virtual work and the applied forces are conservative. We have:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right) - \frac{\partial L}{\partial q_j} = 0$$

Where the lagrangian $L = T - V$.